

Vaccination Data Analysis and Visualization

Public Health and Epidemiology

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Project Category	DS / AI / ML
Tools	Python, MySQL, SQL, Power BI
Dataset Period	2015 - 2025
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End-to-end analytics project covering data cleaning, EDA, relational database design, SQL analysis, and interactive Power BI reporting.

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1. Executive Summary

This project analyzes synthetic vaccination and disease data to understand vaccination coverage, disease incidence, reported disease burden, vaccine introduction, dose schedules, and regional disparities across 27 countries and WHO regions from 2015 to 2025.

The solution follows a complete analytics workflow. Python was used for data inspection, cleaning, validation, exploratory data analysis, and correlation analysis. MySQL was used to build a normalized relational database with primary and foreign keys. SQL queries were used to answer analytical and scenario-based questions. Power BI was connected directly to the MySQL database to build interactive dashboards for vaccination and disease monitoring.

The analysis identifies a temporary decline in vaccination coverage around 2020, a general recovery in later years, differences in coverage across antigens and WHO regions, variation in dose completion, declining disease incidence over time, and an overall vaccine introduction rate of approximately 86.62%. The dataset is synthetic educational data and is therefore interpreted as coursework evidence rather than official public-health statistics.

2. Problem Statement and Objectives

The project requirement is to analyze vaccination and disease data, clean and organize the datasets, store the final data in a structured SQL database, and create interactive Power BI reports that communicate key insights, trends, and comparisons.

Business Objectives

- Identify countries and WHO regions with low vaccination coverage.
- Analyze vaccination coverage and disease incidence trends over time.
- Evaluate dose drop-off between initial and subsequent vaccine doses.
- Compare reported disease burden across diseases.
- Analyze differences in vaccine introduction across regions and years.
- Identify locations that should be prioritized for resource allocation.
- Compare vaccination coverage with disease incidence for related vaccine-disease pairs.
- Provide decision-support insights through interactive Power BI dashboards.

3. Dataset Overview

The project uses five core analytical datasets plus a supporting data dictionary. Shared ISO-3 country codes and year fields allow cross-table joins and consistent analysis.

Dataset	Rows	Purpose	Key Fields
Coverage Data	2,079	Vaccination coverage by country, year and antigen	Code, Year, Antigen, Target number, Doses, Coverage
Incidence Rate	1,782	Disease incidence by country and year	Code, Year, Disease, Denominator, Incidence rate
Reported Cases	1,782	Reported disease case counts	Code, Year, Disease, Cases
Vaccine Introduction	1,188	Vaccine introduction status by country/region	ISO-3 code, WHO region, Year, Description, Intro
Vaccine Schedule	2,258	Vaccination schedule and target population details	ISO-3 code, Year, Vaccine code, Schedule rounds, Target population

Dataset Note

Important: The supplied datasets are synthetic educational data created for coursework. They are not official WHO records and should not be presented as real-world public-health statistics.

4. Methodology and Data Preparation

Step	Phase	Description
1	Data Loading	Loaded all five datasets into Python using Pandas.
2	Data Quality Checks	Reviewed shape, columns, data types, missing values, duplicates, numeric ranges, and year consistency.
3	Cleaning and Standardization	Removed unnecessary whitespace, validated numeric ranges, and confirmed consistent country/year keys.
4	EDA	Created trend, comparison, regional, dose drop-off, and correlation analyses.
5	SQL Modeling	Created a normalized MySQL database with a country dimension and five analytical tables.
6	Power BI Modeling	Connected Power BI to MySQL, created Country and Year relationships, DAX measures, KPI cards, charts, and slicers.
7	Documentation	Compiled the process, Q&A, findings, dashboard screenshots, recommendations, limitations, and conclusions in this report.

5. Python EDA and Findings

5.1 Data Quality Results

- No missing values were found in the supplied core datasets.
- No exact duplicate rows were found.
- Vaccination coverage values were validated to remain between 0 and 100.
- Target population, doses administered, incidence rates, and case counts were checked for non-negative values.
- Country identifiers and annual observations were consistent enough for SQL and Power BI joins.

5.2 Main EDA Insights

Analysis	Finding
Coverage Trend	Average vaccination coverage generally improves over time, with a clear decline around 2020 followed by recovery in later years.
Coverage by Antigen	Coverage differs across antigens, indicating variation in

	uptake and completion across vaccination programs.
Disease Incidence	Average disease incidence generally declines across the study period.
Reported Cases	Tuberculosis contributes the largest reported case burden in the supplied synthetic dataset.
Dose Drop-off	DTP1 vs DTP3 and MCV1 vs MCV2 comparisons show the extent to which later-dose coverage falls below first-dose coverage.
Regional Disparities	Average vaccination coverage varies across WHO regions, supporting targeted resource allocation.
Coverage vs Incidence	Related vaccine-disease pairs generally show an inverse relationship, although correlation is descriptive and does not prove causation.

6. SQL Database Design and Analysis

A MySQL database named `vaccination_analysis` was created. The `countries` table acts as the dimension table and connects to the five analytical fact-style tables through `country_code`. The model uses primary keys, foreign keys, and analytical indexes to support efficient querying.

Table	Final Row Count	Role
<code>countries</code>	27	Country dimension
<code>coverage_data</code>	2,079	Vaccination coverage
<code>incidence_rate</code>	1,782	Disease incidence
<code>reported_cases</code>	1,782	Disease case burden
<code>vaccine_introduction</code>	1,188	Vaccine introduction status
<code>vaccine_schedule</code>	2,258	Schedule and target population

6.1 SQL Analysis Completed

- Countries with the lowest average vaccination coverage.
- Average vaccination coverage by year.
- Total reported cases by disease.
- DTP1-to-DTP3 vaccination drop-off.
- Average vaccination coverage by WHO region.
- Vaccine introduction rate by WHO region and year.
- Measles vaccination coverage compared with measles incidence.
- Country-year observations with high vaccination coverage and high disease incidence.
- Countries below the 95% measles coverage target.
- Resource-allocation prioritization based on low vaccination coverage.
- Measles cases before and after vaccine introduction.

7. Power BI Data Model and Measures

Power BI was connected directly to the MySQL `vaccination_analysis` database. Six SQL tables were loaded. The `countries` table provides country and WHO-region filtering, while a dedicated `Year_Table` dimension provides a consistent year slicer across the analytical tables.

DAX Measures

Measure	Purpose
Average Coverage	Average vaccination coverage percentage.
Total Reported Cases	Total disease cases across the selected filter context.
Average Incidence Rate	Average disease incidence rate.
Total Countries	Distinct number of countries.
Total Doses Administered	Sum of doses administered.
Coverage Gap to 95	Difference between 95% target and average coverage.
Vaccine Introduction Rate	Percentage of introduction records marked Yes.

8. Dashboard - Overview

The Overview page summarizes the most important vaccination and disease indicators. It includes four KPI cards, yearly vaccination coverage, disease burden by disease, WHO-region coverage comparison, and interactive year/region slicers.

Average Coverage	Average Incidence Rate	Total Reported Cases	Total Countries
78.85%	19.86	~53M	27

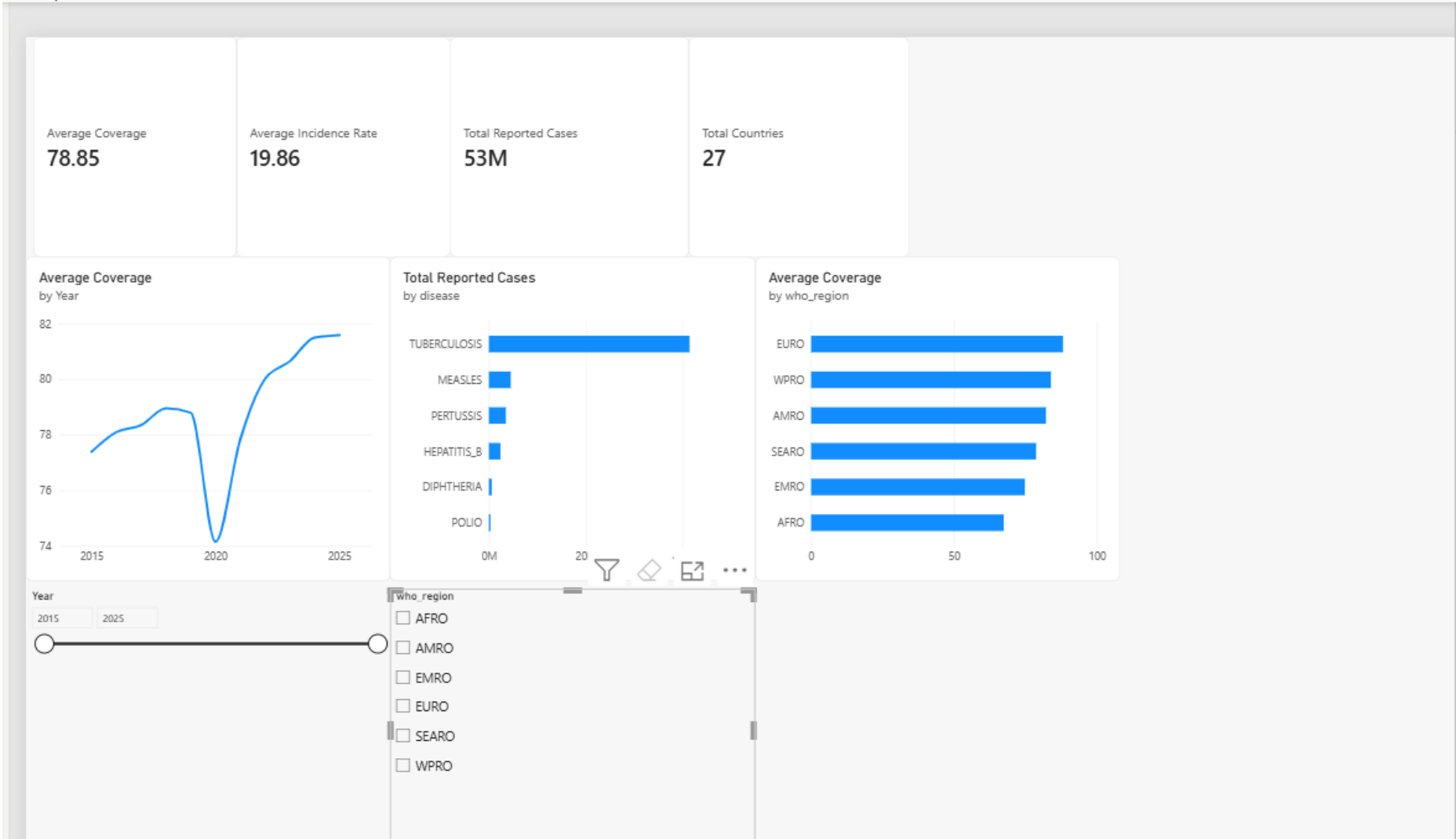


Figure 1. Power BI Overview Dashboard

9. Dashboard - Vaccination Analysis

The Vaccination Analysis page provides deeper analysis of vaccination performance and disease trends. It compares coverage by antigen and country, tracks disease incidence over time, reports the vaccine introduction rate, and provides year and antigen filters.

Vaccine Introduction Rate	Coverage Dimension	Trend Metric
86.62%	Antigen and Country	Average Incidence Rate

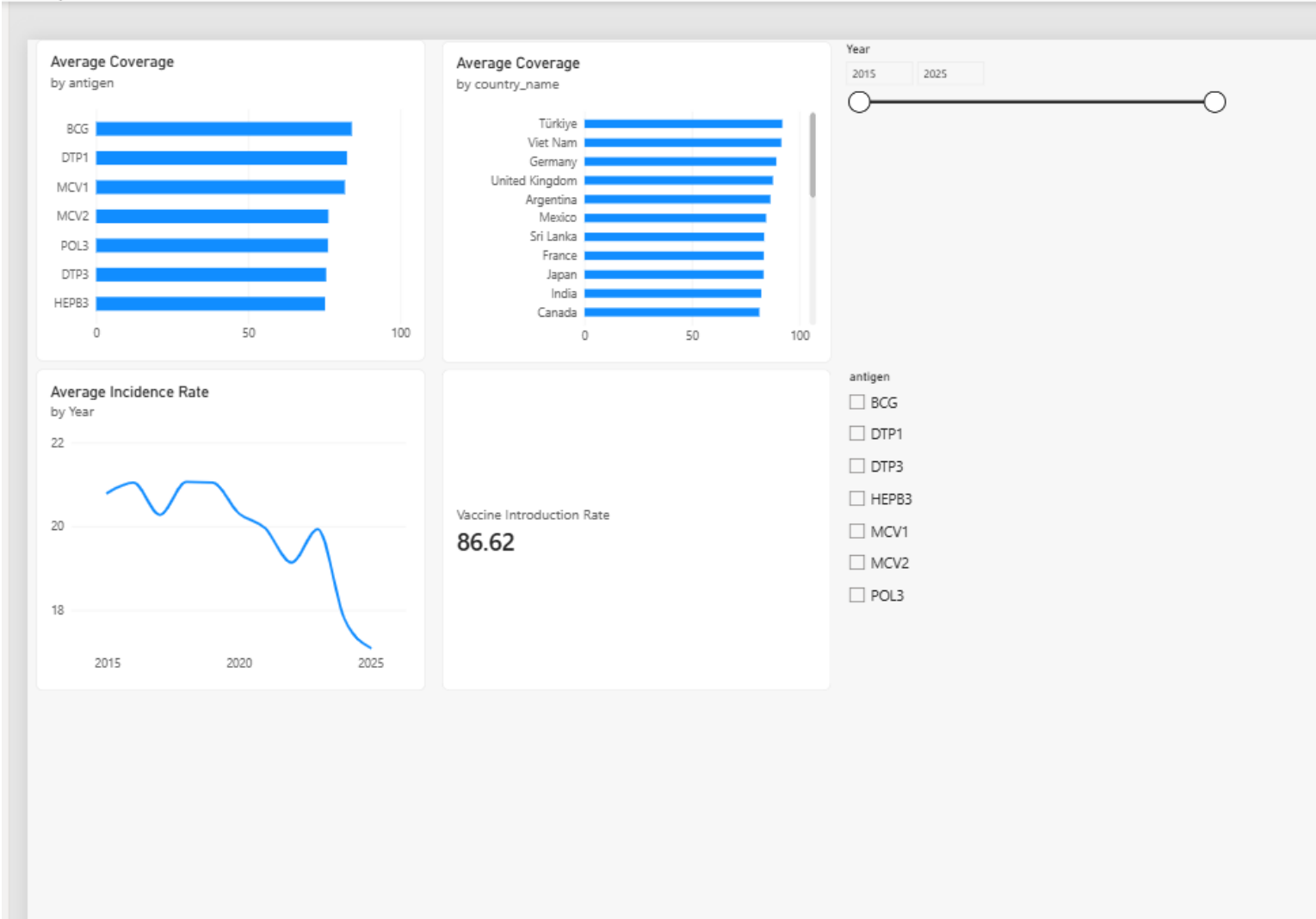


Figure 2. Power BI Vaccination Analysis Dashboard

10. Selected Project Q&A

Q1. How do vaccination rates correlate with a decrease in disease incidence?

Matched vaccine-disease comparisons show that higher coverage is generally associated with lower disease incidence in the synthetic dataset. The relationship is descriptive and does not establish causality.

Q2. What is the drop-off rate between the first dose and subsequent doses?

Dose drop-off was analyzed using DTP1 vs DTP3 and MCV1 vs MCV2. Positive percentage-point differences indicate lower completion of later vaccine doses.

Q3. Are there significant disparities in vaccine introduction timelines across WHO regions?

Yes. Vaccine introduction rates vary across regions and years, indicating differences in the timing and pace of vaccine adoption.

Q4. Which regions or countries should be prioritized for vaccination resources?

Countries and regions with the lowest average coverage should be prioritized for targeted vaccine supply, outreach, and follow-up programs.

Q5. What is the trend in disease cases before and after vaccine introduction?

The before-and-after measles comparison evaluates whether average reported cases fall after introduction. A lower post-introduction average suggests improved disease control within the synthetic dataset.

Q6. Which questions were not answered?

Questions requiring gender, education, socioeconomic status, population density, vaccination strategy type, or true monthly seasonality were not answered because those variables are absent from the supplied datasets.

11. Key Findings and Recommendations

Key Findings

- Average vaccination coverage across the full dataset is approximately 78.85%.
- Vaccination coverage generally increases over time but shows a clear temporary decline around 2020.
- Average disease incidence generally declines over the study period.
- Tuberculosis has the highest total reported case burden in the supplied synthetic dataset.
- Coverage differs by antigen, country, and WHO region.
- Dose drop-off is visible between initial and subsequent doses.

- The overall vaccine introduction rate is approximately 86.62%.
- Low-coverage countries and regions can be identified for targeted resource allocation.

Recommendations

- Prioritize vaccine supply and outreach in low-coverage countries and regions.
- Strengthen follow-up mechanisms for multi-dose vaccination schedules.
- Monitor high-burden diseases and locations with high incidence despite stronger vaccination coverage.
- Track progress toward vaccination targets such as 95% measles coverage.
- Use the Power BI dashboards for ongoing country, region, antigen, and year comparisons.
- If official data becomes available, repeat the same workflow using validated public-health sources before making real policy decisions.

12. Limitations

- The datasets are synthetic educational data and are not official WHO records.
- Gender, education level, socioeconomic status, population density, vaccination strategy type, and true monthly vaccination dates are not available.
- Correlation between vaccination coverage and disease incidence should not be interpreted as proof of causation.
- Before-and-after comparisons do not control for all confounding factors.
- Results are intended for project demonstration and analytical learning.

13. Conclusion

The Vaccination Data Analysis and Visualization project demonstrates a complete analytics pipeline using Python, MySQL, SQL, and Power BI. Python provided data validation and exploratory analysis, MySQL provided structured storage and relational integrity, SQL answered analytical and scenario-based questions, and Power BI transformed the database into interactive decision-support dashboards.

The final solution highlights vaccination trends, disease burden, regional disparities, dose completion gaps, vaccine introduction patterns, and resource-allocation priorities. The project therefore meets the intended coursework deliverables of cleaned and normalized data, a structured SQL database, actionable analysis, interactive Power BI reporting, and documented findings.

Appendix A - SQL Row Validation and Selected Queries

A.1 Final Row Validation

```
SELECT 'countries' AS table_name, COUNT(*) AS row_count FROM countries
UNION ALL
SELECT 'coverage_data', COUNT(*) FROM coverage_data
UNION ALL
SELECT 'incidence_rate', COUNT(*) FROM incidence_rate
UNION ALL
SELECT 'reported_cases', COUNT(*) FROM reported_cases
UNION ALL
SELECT 'vaccine_introduction', COUNT(*) FROM vaccine_introduction
UNION ALL
SELECT 'vaccine_schedule', COUNT(*) FROM vaccine_schedule;
```

A.2 Lowest-Coverage Countries

```
SELECT c.country_name, ROUND(AVG(cd.coverage), 2) AS avg_coverage
FROM coverage_data cd
JOIN countries c ON cd.country_code = c.country_code
GROUP BY c.country_name
ORDER BY avg_coverage ASC
LIMIT 10;
```

A.3 Average Coverage by WHO Region

```
SELECT c.who_region, ROUND(AVG(cd.coverage), 2) AS avg_coverage
FROM coverage_data cd
JOIN countries c ON cd.country_code = c.country_code
GROUP BY c.who_region
ORDER BY avg_coverage DESC;
```

A.4 Measles Coverage Gap to 95%

```
SELECT c.country_name, cd.year,
       ROUND(AVG(cd.coverage), 2) AS measles_coverage,
       ROUND(95 - AVG(cd.coverage), 2) AS gap_to_95_percent
FROM coverage_data cd
JOIN countries c ON cd.country_code = c.country_code
WHERE cd.antigen = 'MCV1'
GROUP BY c.country_name, cd.year
HAVING AVG(cd.coverage) < 95
ORDER BY gap_to_95_percent DESC;
```